Photosynthesis

Food from light

Green plants make their own food. In photosynthesis, a plant uses the energy of sunlight to turn carbon dioxide from the air and water from the soil into glucose, a sugar, and gives out oxygen. The overall equation is: six carbon dioxide plus six water, with light and chlorophyll, gives one glucose plus six oxygen.

Where it happens

Photosynthesis takes place in chloroplasts, small green bodies inside the cells of leaves. Chloroplasts contain chlorophyll, a green pigment that absorbs red and blue light and reflects green light, which is why leaves look green. Carbon dioxide enters the leaf through tiny pores called stomata, usually on the underside of the leaf. Water travels up from the roots through tubes called xylem.

Two stages

The light reactions happen in the thylakoid membranes. Light energy splits water molecules, releasing oxygen, and stores energy in two carrier molecules, ATP and NADPH. The Calvin cycle happens in the stroma, the fluid around the thylakoids. It uses ATP and NADPH to fix carbon dioxide into sugar. It does not need light directly, but it needs the products of the light reactions.

What limits it

The rate of photosynthesis depends on light intensity, carbon dioxide concentration and temperature. If one of these is in short supply, it limits the rate even when the others are plentiful. In a greenhouse, farmers raise carbon dioxide to about 1000 parts per million, more than twice the air's level, to grow tomatoes faster.

Why it matters

Almost all food chains begin with photosynthesis, and it produces the oxygen we breathe. Forests and ocean algae together absorb roughly a quarter of the carbon dioxide that humans release each year.